Circles

Solutions

Starter

Solution 2

1. Complete the square: write each expression as $(x+p)^2+q$.

1(a) $x^2+10x+28$

1(a) $(x+5)^2+3$

1(b) x^2-6x+1

1(b) $(x-3)^2-8$

1(c) x^2-12x

1(c) $(x-6)^2-36$

1(d) x^2+7x

1(d) $(x+\frac{7}{2})^2-\frac{49}{4}$

2. Find the line through each pair of points.

2(a) A(0,−6), B(4,3)

2(a) $y=\frac{9}{4}x-6$

2(b) P(7,−5), Q(−9,3)

2(b) $y=-\frac{1}{2}x-\frac{3}{2}$

2(c) R(−4,−2), T(5,10)

2(c) $y=\frac{4}{3}x+\frac{10}{3}$

3. Use the discriminant to determine the number of real solutions.

3(a) $x^2-7x+14=0$

3(a) $49-56=-7$

No real solutions.

3(b) $x^2+11x+8=0$

3(b) $121-32=89$

Two real solutions.

3(c) $4x^2+12x+9=0$

3(c) $144-144=0$

One repeated real solution.

4 Find the line through (3,−4) perpendicular to $6x-5y-1=0$.

4 $y+4=-\frac{5}{6}(x-3) \Rightarrow y=-\frac{5}{6}x-\frac{3}{2}$

Chapter Overview

Solution 3

1. Circle equations

Find centres, radii and equations.

2. Lines and circles

Find intersections and identify tangency.

3. Chords and tangents

Use perpendicular bisectors and circle geometry.

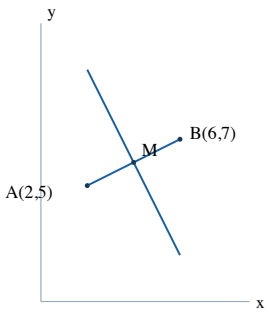
4. Triangles in circles

Find the circle through three vertices.

Midpoints and Perpendicular Bisectors

Solution 4

A perpendicular bisector passes through the midpoint of a segment and meets it at a right angle.



1 Find the midpoint of A(2,5) and B(6,7).

1

$$M = (\frac{2+6}{2}, \frac{5+7}{2}) = (4, 6)$$

2 Find the gradient of the perpendicular bisector.

2

$$m_{AB} = \frac{7-5}{6-2} = \frac{1}{2}$$

$$m_{\perp} = -2$$

3 Find its equation.

3

$$y - 6 = -2(x - 4)$$

$$y = -2x + 14$$

The perpendicular bisector of a chord passes through the centre of its circle.

Test Your Understanding — Solution 5

a Find the perpendicular bisector of A(4,7), B(10,17).

a

$M = (7, 12), \quad m_{AB} = \frac{5}{3}$

$y - 12 = -\frac{3}{5}(x - 7)$

b Find a general perpendicular-bisector equation for distinct points $A(x_1, y_1), B(x_2, y_2)$.

b

$(x_2 - x_1)(x - \frac{x_1 + x_2}{2})$

$+(y_2 - y_1)(y - \frac{y_1 + y_2}{2}) = 0$

This also covers horizontal and vertical segments.

c AB is a diameter with centre (5,−4). A=(1,−2). Find B.

c

$B = 2(5, -4) - (1, -2) = (9, -6)$

Derive the bisector from its midpoint and perpendicular direction; there is no need to memorise the general expression.

Exercise 6A - Y1 Page 115 — Solution 6

6 JK is a diameter with $J(\frac{3}{4}, \frac{4}{3})$, $K(-\frac{1}{2}, 2)$. The centre lies on $y = 8x + b$. Find b.

6

$C = (\frac{1}{8}, \frac{5}{3})$

$\frac{5}{3} = 8(\frac{1}{8}) + b \Rightarrow b = \frac{2}{3}$

7 AB is a diameter with A(0,−2), B(6,−5). Show its centre lies on x−2y−10=0.

7

$C = (3, -\frac{7}{2})$

$3 - 2(-\frac{7}{2}) - 10 = 0$

Exercise 6B - Y1 Page 116

Solution 7

2 F(−2,5), G(2,9) are diameter endpoints. Find the line through the centre perpendicular to FG.

2

$C = (0, 7), \quad m_{FG} = 1$

$y - 7 = -(x - 0)$

$y = -x + 7$

3 J(0,−3), K(4,−5) are diameter endpoints. Find the perpendicular line through the centre in integer standard form.

3

$P = (2, -4), \quad m_{JK} = -\frac{1}{2}$

$y + 4 = 2(x - 2)$

$2x - y - 8 = 0$

Exercise 6B - Y1 Page 117

Solution 8

5 X=(7,−2), Y=(4,q). Their perpendicular bisector is y=4x+b. Find q and b.

5

$\frac{q + 2}{-3} = -\frac{1}{4} \Rightarrow q = -\frac{5}{4}$

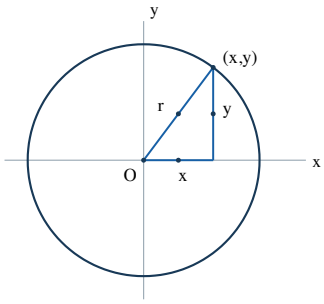
$M = (\frac{11}{2}, -\frac{13}{8})$

$b = -\frac{13}{8} - 4(\frac{11}{2}) = -\frac{189}{8}$

Equation of a circle

Solution 9

A point (x, y) is on a circle centred at the origin with radius r. Draw a right-angled triangle from the origin to the point.



Equation What equation must x and y satisfy?

Equation

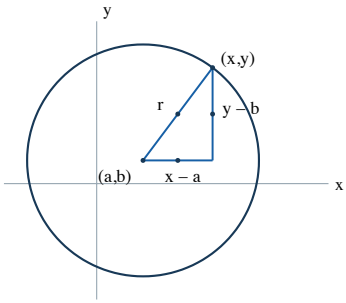
$x^2 + y^2 = r^2$

By Pythagoras: the horizontal and vertical distances are x and y, and the hypotenuse is r.

Equation of a circle

Solution 10

The centre moves to (a, b). The horizontal and vertical displacements to (x, y) are $x - a$ and $y - b$.



Equation Use Pythagoras to find the equation.

Equation

$(x - a)^2 + (y - b)^2 = r^2$

Quickfire Questions

Solution 11

Find the missing equation, or the centre and radius, for each circle.

a Centre

$(0, 0)$

Radius

5

a $x^2 + y^2 = 25$

b Centre

$(1, 2)$

Radius

6

b $(x - 1)^2 + (y - 2)^2 = 36$

c

$(x + 3)^2 + (y - 5)^2 = 1$

c

Centre

$(-3, 5)$

Radius

1

d

$(x + 5)^2 + (y - 2)^2 = 49$

d

Centre

$(-5, 2)$

Radius

7

e

$(x + 6)^2 + y^2 = 16$

e

Centre

$(-6, 0)$

Radius

4

f

$(x - 1)^2 + (y + 1)^2 = 3$

f

Centre

$(1, -1)$

Radius

$\sqrt{3}$

g

$(x + 2)^2 + (y - 3)^2 = 8$

g

Centre

$(-2, 3)$

Radius

$2\sqrt{2}$

Finding the equation using points

Solution 12

AB is a diameter with A(5,8), B(−7,4). Find the circle equation.

1 Find the centre.

1

$C = (-1, 6)$

2 Find the squared radius.

2

$r^2 = (5 + 1)^2 + (8 - 6)^2 = 40$

3 Write the equation.

3

$(x + 1)^2 + (y - 6)^2 = 40$

Test Your Understanding

Solution 13

A=(5,−1), B=(13,11). AB is a diameter of circle C.

a Find the midpoint of AB.

a

$M = (9, 5)$

b Find an equation for C.

b

$r^2 = 4^2 + 6^2 = 52$

$(x-9)^2 + (y-5)^2 = 52$

Completing the square

Solution 14

Find the centre and radius of

$x^2 + y^2 - 6x + 2y - 6 = 0$

1 Group the x and y terms, then complete both squares.

1

$(x-3)^2 - 9 + (y+1)^2 - 1 - 6 = 0$

2 Read the centre and radius from the standard form.

2

$(x-3)^2 + (y+1)^2 = 16$

$C = (3, -1), \quad r = 4$

Further Example

Solution 15

A circle has equation

$x^2 + y^2 - 20x - 16y + 139 = 0$

a Find its centre.

b Show its radius is 5.

a

$(x - 10)^2 + (y - 8)^2 = 25$

$T = (10, 8)$

b

$r = \sqrt{25} = 5$

Example

Solution 16

Derive the centre and radius of

$x^2 + y^2 + ax + by + c = 0$

1 Complete the square.

1

$(x + \frac{a}{2})^2 + (y + \frac{b}{2})^2 = \frac{a^2 + b^2}{4} - c$

2 Identify the centre and radius.

2

$C = (-\frac{a}{2}, -\frac{b}{2})$

$r = \sqrt{\frac{a^2 + b^2}{4} - c}$

A real circle requires the expression under the square root to be positive. Zero gives a single point; a negative value gives no real points.

Exercise 6C - Y1 Page 120

Solution 17

5 PQ is a diameter, with P(5,6) and Q(−2,2). Find the circle equation.

5

$C = (\frac{3}{2}, 4), \quad r^2 = (5 - \frac{3}{2})^2 + (6 - 4)^2 = \frac{65}{4}$

$(x - \frac{3}{2})^2 + (y - 4)^2 = \frac{65}{4}$

6 The point $(1,-3)$ lies on $(x-3)^2 + (y+4)^2 = r^2$. Find r.

6

$r^2 = (1 - 3)^2 + (-3 + 4)^2 = 5$

$r = \sqrt{5}$

Exercise 6C - Y1 Page 120

Solution 18

Complete the square, then find each centre and radius.

8(a)

$x^2 + y^2 - 4x - 11 = 0$

8(a)

$(x - 2)^2 + y^2 = 15$

9(a)

$x^2 + y^2 - 10x + 4y - 20 = 0$

9(a)

$(x - 5)^2 + (y + 2)^2 = 49$

8(b) Read the centre and radius.

8(b)

$C = (2, 0), \quad r = \sqrt{15}$

9(b) Read the centre and radius.

9(b)

$C = (5, -2), \quad r = 7$

Exercise 6C - Y1 Page 120

Solution 19

12 P(7,−14) is on $x^2+y^2+6x-14y=483$. PQ is a diameter. Find Q.

12

$(x+3)^2+(y-7)^2=541$

$C=(-3,7)$

$Q=2C-P=(-13,28)$

13 The circle $(x-k)^2+y^2=41$ passes through (3,4). Find both values of k.

13

$(3-k)^2+16=41$

$(3-k)^2=25$

$k=-2$ or 8

Extension

Solution 20

1 [MAT 2009 1B] Find the shortest distance from the origin to $x^2+y^2+6x+8y=75$.

1

$(x+3)^2+(y+4)^2=100$

$C=(-3,-4)$

$r=10, \quad OC=5$

The origin is inside the circle. The nearest point is on the ray from C through O.

$d=10-5=5$

2 [MAT 2016 1I] For positive a,b and $x^2+y^2\le 1$, find the greatest possible value of ax+by.

2

The parallel lines $ax+by=k$ sweep across the unit disk. The maximum occurs when the line is tangent, at distance 1 from the origin.

$\frac{k}{\sqrt{a^2+b^2}}=1$

$k=\sqrt{a^2+b^2}$

3 [MAT 2007 1D] Which point on $(x-5)^2+(y-4)^2=4$ is closest to $(x-1)^2+(y-1)^2=1$?

3

The centres are 5 apart. Move $\frac{2}{5}$ of the way from (5,4) towards (1,1).

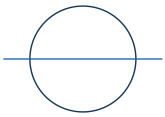
$P=(5,4)+\frac{2}{5}(-4,-3)$

$P=(\frac{17}{5},\frac{14}{5})$

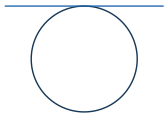
Intersections of Lines and Circles

Solution 21

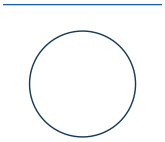
Solve a line and circle simultaneously by substitution. The resulting quadratic has two, one or no real roots: a secant, a tangent or no intersection.



Secant: two intersections



Tangent: one intersection



No intersection

Example Show that $y = x + 3$ never intersects $x^2 + y^2 = 1$.

Example

$x^2 + (x+3)^2 = 1$

$x^2 + 3x + 4 = 0$

$D = 3^2 - 4(1)(4) = -7 < 0$

There are no real solutions, so no points of intersection.

Test Your Understanding

Solution 22

1 Find where $y = x + 6$ meets $x^2 + (y-3)^2 = 29$.

1

$x^2 + (x+3)^2 = 29$

$x^2 + 3x - 10 = 0$

$(x+5)(x-2) = 0$

$(-5, 1), \quad (2, 8)$

2 Use algebra to find k when $y = x + k$ touches $x^2 + y^2 = 1$.

2

$2x^2 + 2kx + k^2 - 1 = 0$

Tangency means a repeated root, so the discriminant is zero.

$4k^2 - 8(k^2 - 1) = 0$

$k = \pm\sqrt{2}$

Exercise 6D - Y1 Page 122 — Solution 23

6. The line $x + y = 11$ meets the circle $x^2 + (y-3)^2 = 32$.

a Show that there is only one intersection.

a

$y = 11 - x$

$x^2 + (8-x)^2 = 32$

$x^2 - 8x + 16 = 0$

$(x-4)^2 = 0$

One repeated root: the line is tangent.

b Find the point of intersection.

b

$x = 4, \quad y = 11 - 4 = 7$

$(4, 7)$

Exercise 6D - Y1 Page 122 — Solution 24

7. The line $y = 2x - 2$ meets $(x-2)^2 + (y-2)^2 = 20$ at A and B.

7(a) Find A and B.

7(a)

$(x-2)^2 + (2x-4)^2 = 20$

$5(x-2)^2 = 20$

$x = 0$ or 4

$A = (0, -2), \quad B = (4, 6)$

7(b) Show that AB is a diameter.

7(b)

$M_{AB} = (\frac{0+4}{2}, \frac{-2+6}{2}) = (2, 2)$

The midpoint is the centre of the circle, so AB is a diameter.

11 Find the possible k for which $y = 4x - 1$ does not meet the circle $x^2 + 2x + y^2 = k$.

11

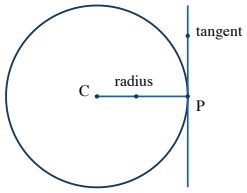
$17x^2 - 6x + 1 - k = 0$

$D = 36 - 68(1-k) < 0 \Rightarrow k < \frac{8}{17}$

A real circle also needs $k + 1 > 0$, so $-1 < k < \frac{8}{17}$.

Tangents, Chords, Perpendicular Bisectors

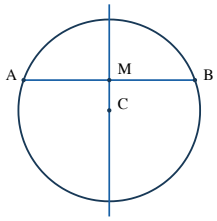
Solution 25



Tangent and radius

The tangent at a point on a circle is perpendicular to the radius to that point.

Find the radius gradient, take its negative reciprocal, then use the point to find the tangent equation.



Chord and centre

The perpendicular bisector of any chord passes through the centre.

Find the perpendicular bisectors of two chords. Their intersection is the circle centre.

Examples

Solution 26

1. Circle C: $(x-3)^2+(y-7)^2=100$, with P(11,1).

1(a) Verify P is on C.

1(a) $(11-3)^2+(1-7)^2=64+36=100$

1(b) Find the tangent at P.

1(b)

$m_r=-\frac{3}{4}, \quad m_t=\frac{4}{3}$

$y-1=\frac{4}{3}(x-11)$

$4x-3y-41=0$

2. Find both tangents of gradient -3 to $(x-4)^2+(y+4)^2=10$.

2

The radius to a contact point has gradient $\frac{1}{3}$.

$y+4=\frac{1}{3}(x-4)$

Substitution into the circle gives contact points $(1,-5)$ and $(7,-3)$.

$y+5=-3(x-1)$

$y=-3x-2$

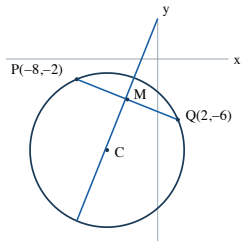
$y+3=-3(x-7)$

$y=-3x+18$

Determining the Circle Centre

Solution 27

P(−8,−2) and Q(2,−6) lie on a circle with centre C. M is the midpoint of PQ. The line l passes through M and C.



a Find an equation of l.

a

$$M = (-3,-4), \quad m_{PQ} = -\frac{2}{5}$$

$$l:\ y+4 = \frac{5}{2}(x+3)$$

b(i) Given that the y-coordinate of C is −9, show its x-coordinate is −5.

b(i)

$$-9+4 = \frac{5}{2}(x+3) \Rightarrow x = -5$$

b(ii) Find the circle equation.

b(ii)

$$C = (-5,-9)$$

$$r^2 = 3^2 + 7^2 = 58$$

$$(x+5)^2 + (y+9)^2 = 58$$

Test Your Understanding

Solution 28

1 Find the tangent at P(6,9) to the circle with centre C(3,5). Use integer standard form.

1

$$m_r = \frac{4}{3}, \quad m_t = -\frac{3}{4}$$

$$y-9 = -\frac{3}{4}(x-6)$$

$$3x+4y-54 = 0$$

2 A circle passes through A(0,0), B(4,2). Its centre has x-coordinate −1. Find its equation.

2

$$M = (2,1), \quad m_{AB} = \frac{1}{2}$$

$$y-1 = -2(x-2)$$

$$x = -1 \Rightarrow y = 7$$

$$r^2 = 1^2 + 7^2 = 50$$

$$(x+1)^2 + (y-7)^2 = 50$$

Exercise 6E - Y1 Page 126-128 — Solution 29

6. The tangent to $(x+4)^2+(y-1)^2=242$ at $(7,-10)$ meets the y-axis at S and the x-axis at T.

a Find S and T.

a

$C=(-4,1), \quad m_r=\dfrac{-11}{11}=-1$

$m_t=1, \quad y+10=x-7$

$y=x-17$

$S=(0,-17), \quad T=(17,0)$

b Find the area of triangle OST, where O is the origin.

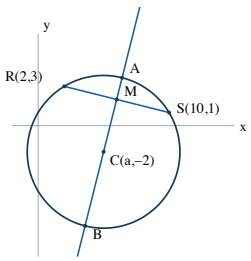
b

$\text{Area}=\dfrac{1}{2}\times 17\times 17=\dfrac{289}{2}$

The area is 144.5 square units.

Exercise 6E - Y1 Page 126-128 — Solution 30

10. R(2,3), S(10,1) lie on a circle with centre C(a,−2). M is the midpoint of RS. The line l passes through M and C.



a Find l.

a

$M=(6,2), \quad m_{RS}=-\dfrac{1}{4}$

$y-2=4(x-6)$

$y=4x-22$

b Find a.

b

$-2=4a-22$

$a=5$

c Find the circle equation.

c

$r^2=(2-5)^2+(3+2)^2=34$

$(x-5)^2+(y+2)^2=34$

d Find where l meets the circle.

d

$(x-5)^2+(4x-20)^2=34$

$(x-5)^2=2 \Rightarrow x=5\pm\sqrt{2}$

$(5+\sqrt{2},-2+4\sqrt{2})$

$(5-\sqrt{2},-2-4\sqrt{2})$

Exercise 6E - Y1 Page 126-128

Solution 31

1 [MAT 2012 1A] Which line is tangent to $x^2 + y^2 = 4$?

A: $x + y = 2$

B: $y = x - 2\sqrt{2}$

C: $x = \sqrt{2}$

D: $y = \sqrt{2} - x$

1

B. Its distance from the origin is $2\sqrt{2}/\sqrt{2} = 2$, equal to the radius.

2. [**AEA 2006 Q4**]

$y = mx$ is tangent to $(x + 4)^2 + (y - 7)^2 = 13$.

(a) Show $3m^2 + 56m + 36 = 0$.

(b) Find the two contact points A and B.

(c) Tangents from (4,−7) touch $x^2 + y^2 = 13$ at P and Q. Find P or Q.

Worked solution: next slide.

3. [**STEP 2005 Q6**]

A=(5,16), B=(−4,4), and AP=2BP.

(i) Show the locus of P is $(x + 7)^2 + y^2 = 100$.

(ii) C=(a,0), D=(b,0), QC=kQD, k>1. This gives the same locus.

Show $\dfrac{a+7}{b+7} = \dfrac{a^2+51}{b^2+51}$.

For a≠b, show $(a + 7)(b + 7) = 100$.

Worked solution: slide 33.

Mark Scheme for Extension Question 2

Solution 32

[AEA 2006 Q4] Tangents from the origin to $(x + 4)^2 + (y - 7)^2 = 13$.

a Find the gradient condition.

b Find the contact points.

a

$(1 + m^2)x^2 + (8 - 14m)x + 52 = 0$

$(8 - 14m)^2 - 208(1 + m^2) = 0$

$3m^2 + 56m + 36 = 0$

b

$(3m + 2)(m + 18) = 0$

$m = -\dfrac{2}{3}$ or -18

$x = \dfrac{14m - 8}{2(1 + m^2)}$

$A = (-6, 4), \quad B = (-\dfrac{2}{5}, \dfrac{36}{5})$

c Find a contact point for tangents from (4,−7) to $x^2 + y^2 = 13$.

c

Translate the previous configuration by (4,−7). The circle centre moves to the origin.

$P = (-6, 4) + (4, -7) = (-2, -3)$

$Q = (-\dfrac{2}{5}, \dfrac{36}{5}) + (4, -7) = (\dfrac{18}{5}, \dfrac{1}{5})$

Mark Scheme for Extension Question 3 — Solution 33

[STEP 2005 Q6] A=(5,16), B=(−4,4), AP=2BP.

i Find the locus.

i

$$(x-5)^2+(y-16)^2=4((x+4)^2+(y-4)^2)$$

$$3x^2+3y^2+42x-153=0$$

$$(x+7)^2+y^2=100$$

ii Compare the locus QC=kQD with the same circle.

ii

$$(k^2-1)(x^2+y^2)+(2a-2k^2b)x+k^2b^2-a^2=0$$

$$\frac{2a-2k^2b}{k^2-1}=14,\quad \frac{k^2b^2-a^2}{k^2-1}=-51$$

$$k^2=\frac{a+7}{b+7}=\frac{a^2+51}{b^2+51}$$

Conclusion Deduce the result when $a\neq b$.

Conclusion

$$(a+7)(b^2+51)=(b+7)(a^2+51)$$

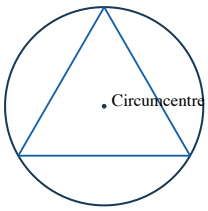
$$(b-a)(ab+7a+7b-51)=0$$

$$(a+7)(b+7)=100$$

Triangles in Circles — Solution 34

A triangle is inscribed in a circle when all three vertices lie on the circle.

The circle circumscribes the triangle. It is called the circumcircle; its centre is the circumcentre.

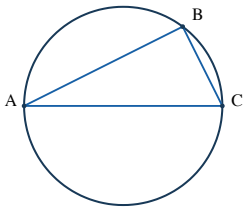


Triangles in Circles

Solution 35

If $\angle ABC = 90°$, AC is the diameter of the circumcircle.

Conversely, if AC is a diameter and B is on the circle, $\angle ABC = 90°$. Thus AB is perpendicular to BC and $AB^2 + BC^2 = AC^2$.



To find the circumcentre of any triangle: find the perpendicular bisectors of two sides, then solve for their intersection.

Example

Solution 36

A=(−8,1), B=(4,5), C=(−4,9).

a: method 1 Show AB is a diameter using Pythagoras.

a: method 1

$AC^2 = 4^2 + 8^2 = 80$

$BC^2 = 8^2 + 4^2 = 80$

$AB^2 = 12^2 + 4^2 = 160$

Since $AC^2 + BC^2 = AB^2$, angle ACB is a right angle. AB is a diameter.

a: method 2 Show the two sides are perpendicular.

a: method 2

$m_{AC} = 2, \quad m_{BC} = -\frac{1}{2}$

The gradients multiply to -1, so $\angle ACB = 90°$.

b Find the circle equation.

b

$O = (-2, 3), \quad r^2 = 6^2 + 2^2 = 40$

$(x + 2)^2 + (y - 3)^2 = 40$

Example

Solution 37

Find the circle through A(0,2), B(2,0) and C(8,18).

1 Find two perpendicular bisectors. **2** Find the centre and radius. **3** Write the circle equation.

1

$M_{AB} = (1,1), \quad m_{AB} = -1$

$y = x$

$M_{AC} = (4,10), \quad m_{AC} = 2$

$y - 10 = -\frac{1}{2}(x-4)$

2

$x - 10 = -\frac{1}{2}(x-4) \Rightarrow x = 8$

$O = (8,8), \quad r^2 = 8^2 + 6^2 = 100$

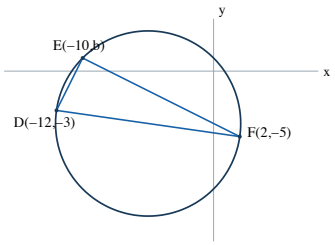
3

$(x-8)^2 + (y-8)^2 = 100$

Exercise 6F - Y1 Page 131-132

Solution 38

8. D(−12,−3), E(−10,b), F(2,−5) lie on a circle. Given $\angle DEF = 90°$ and b>0:



a Show b=1. **b** Find the circle equation.

a

$DE^2 = 4 + (b+3)^2$

$EF^2 = 144 + (b+5)^2, \quad DF^2 = 200$

$DE^2 + EF^2 = DF^2$

$2b^2 + 16b - 18 = 0$

$(b+9)(b-1) = 0$

Since b>0, b=1.

b

DF is a diameter, so its midpoint is the centre.

$O = (-5,-4), \quad r^2 = 200/4 = 50$

$(x+5)^2 + (y+4)^2 = 50$

Exercise 6F - Y1 Page 131-132 Solution 39

9. A circle has equation $x^2 + 2x + y^2 - 24y - 24 = 0$.

a Find its centre and radius.

a

$(x+1)^2 + (y-12)^2 = 169$

$O = (-1, 12), \quad r = 13$

b A(−13,17) and B(11,7) lie on the circle. Show AB is a diameter.

b

$M_{AB} = (-1, 12) = O$

The chord passes through the centre, so it is a diameter.

c C lies on the negative x-axis and angle ACB is a right angle. Find C.

c

C lies on the circle with diameter AB. Put y=0:

$(x+1)^2 + 144 = 169$

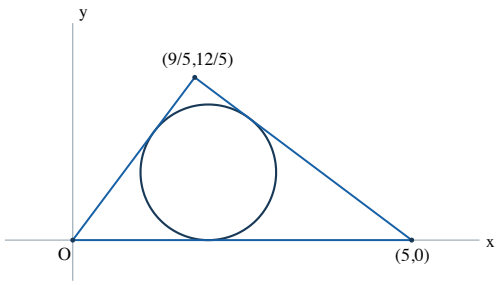
$x + 1 = \pm 5 \Rightarrow x = 4$ or -6

$C = (-6, 0)$

Exercise 6F - Y1 Page 131-132 Solution 40

[STEP] A circle touches all three sides of the triangle bounded by $y = 0$, $3y = 4x$ and $4y + 3x = 15$. Find its equation.

Given: the family $(x - 2t)^2 + (y - t)^2 = t^2$, t>0, touches the first two lines. The incircle lies entirely inside the triangle.



1 Use the first two sides.

1

Tangency to y=0 gives centre (h,t) and radius t>0. For a centre inside the triangle, its distance to $4x - 3y = 0$ gives

$\frac{4h - 3t}{5} = t \Rightarrow h = 2t$

2 Use the third side.

2

The distance from (2t,t) to $3x + 4y = 15$ is t. Inside the triangle,

$\frac{15 - 10t}{5} = t \Rightarrow t = 1$

3 Write the equation.

3

$C = (2, 1), \quad r = 1$

$(x-2)^2 + (y-1)^2 = 1$